Data Analyst Intern Assessment

ASSESSMENT INSTRUCTIONS

Position: Data Analyst Intern

- **Start Time: 04th February 2026, 10:00 AM (IST)**
- **Deadline: 05th February 2026, 10:30 AM (IST)**

STRICT 24-HOUR DEADLINE - NO EXTENSIONS GRANTED

Submission Requirements:

- **File Name:** Data Analyst Intern_[Your_Name] _February2026
- **Email to:** [career@purplemerit.com] with subject "Data Analyst Intern Assessment Submission - [Your Name]"

❖ LATE SUBMISSIONS ARE DISQUALIFIED.

ASSESSMENT OVERVIEW

Urban Threads — D2C Fashion Brand - Order & Returns Analytics Assessment

ASSESSMENT OVERVIEW

Company Context:

Urban Threads is a direct-to-consumer fashion brand that launched in 2021. We sell contemporary workwear and casual apparel through our website and mobile app. Over the past year, we've scaled from 5,000 to 45,000 monthly orders, but our operations team has raised concerns about rising return rates and their impact on profitability.

Role You're Applying For:

Data Analyst Intern — Operations & Customer Analytics Team

What This Assessment Evaluates:

This assessment mirrors the actual work you'd do in your first month. We're not looking for perfection we want to see how you think through messy data, what questions you ask, and whether you can translate numbers into business recommendations.

Time to Complete:

3-4 hours (you may take breaks; just track total working time)

Submission Format:

- One Excel/Google Sheets file with your analysis (clearly labelled tabs)
- One PDF document (2-3 pages max) with your findings, visualizations, and recommendations
- Include a brief write-up of any assumptions you made

Tools Allowed:

- Excel or Google Sheets (required)
- Any visualization tool you're comfortable with (Excel charts, Google Sheets, Tableau Public, Power BI Desktop)
- SQL is not required but you may note any queries you'd write if you had database access

DATASET DESCRIPTION

File Name: urbanthread_orders_q4_2024 (attached in the same email)

Business Background:
This dataset contains all orders placed on Urban Threads' website and app during Q4 2024 (October 1 – December 31). It includes order-level information along with return status for items that were sent back. The Finance team uses this data for revenue reporting, while Operations tracks fulfillment efficiency and return rates.

Important Context:

- Our fiscal month runs from the 1st to the last day of each calendar month
- Return window is 30 days from delivery date
- We operate only in India, with warehouses in Mumbai, Delhi, and Bangalore
- Peak season was Diwali (late October/early November)

Column Descriptions

Column Name	Data Type	Description
order_id	String	Unique identifier for each order (format: UT-XXXXXX)
order_date	Date	Date when the order was placed

Column Name	Data Type	Description
customer_id	String	Unique customer identifier (format: CUST_XXXXX)
customer_type	String	“New” (first order) or “Returning” (2+ historical orders)
product_sku	String	Stock Keeping Unit (format: CAT-STYLE-SIZE, e.g., TOP-CAS-M)
product_category	String	High-level category: Tops, Bottoms, Dresses, Outerwear, Accessories
product_name	String	Full product name
unit_price	Numeric	Price per unit in INR
quantity	Integer	Number of units ordered
order_total	Numeric	Total order value in INR (unit_price × quantity; to be validated)
discount_code	String	Promo code applied; blank if no discount
discount_percent	Numeric	Discount percentage applied (0–100)
payment_method	String	COD, Credit Card, Debit Card, UPI, Wallet
order_source	String	Website, Mobile App, Instagram Shop
shipping_city	String	Customer’s city
shipping_state	String	Customer’s state
warehouse_origin	String	Fulfilment warehouse: Mumbai, Delhi, Bangalore
delivery_date	Date	Actual delivery date (blank if not delivered or cancelled)
delivery_status	String	Delivered, In Transit, Cancelled, RTO (Return to Origin)
return_initiated	String	Yes/No — whether a return was initiated
return_reason	String	Size Issue, Quality Defect, Wrong Item, Changed Mind, Damaged in Transit, Other

Column Name	Data Type	Description
return_received	Date	Date when returned item reached warehouse
refund_status	String	Processed, Pending, Not Applicable

Business Objectives Behind This Data:

1. Understand what's driving our return rate, which jumped from 12% to 19% this quarter
2. Identify whether certain product categories, customer segments, or order sources have disproportionate issues
3. Assess whether our Diwali promotions attracted low-quality orders (high return, high COD)
4. Provide actionable recommendations to reduce returns without hurting sales

SECTION 1: DATA CLEANING & PREPARATION

Before analyzing, you need to ensure the data is reliable. Document every cleaning step in a separate "Cleaning Log" tab.

Tasks:

1.1 Missing Value Analysis

- Identify all columns with missing values
- For each, determine: Is this truly missing data or expected based on business logic?
- Example: A blank return_reason for an order where return_initiated = "No" is expected, not an error
- Document how you handled each type of missing value

1.2 Duplicate Detection

- Check for duplicate order_id values
- If duplicates exist, investigate whether they're true duplicates or data entry errors (same order_id but different product_sku might indicate a multi-item order vs. an actual error)
- State your decision and reasoning

1.3 Format Inconsistencies

- Examine date columns — are all dates in consistent format?
- Check text fields for inconsistent capitalization, trailing spaces, or typos (e.g., "Delievered" vs "Delivered")
- Standardize product_sku format if variations exist

1.4 Outlier Identification

- Flag any orders with unusually high quantity (what's reasonable for a fashion order?)
- Identify unit_price values that seem off (e.g., ₹0, negative, or unreasonably high for apparel)
- Note any discount_percent values outside 0-100 range

1.5 Logical Data Issues

- Cross-check: Does order_total = unit_price × quantity after applying discount_percent?
- Verify: Can delivery_date logically occur before order_date?
- Check: If delivery_status = "Cancelled," should there be a delivery_date?
- Confirm: If return_initiated = "Yes," is there always a return_reason?
- Investigate: Are there any orders with return_received dates before delivery_date?

SECTION 2: DATA VALIDATION & QUALITY CHECKS

Create a "Validation Summary" tab documenting your checks.

Validation Rules to Apply:

Rule ID	**Validation Check**	**Expected Outcome**
V1	All order_id values follow the format **UT-XXXXXX**	100% compliance
V2	customer_type contains only **"New"** or **"Returning"**	No other values
V3	quantity is a positive integer	No zeros, negatives, or decimals

Rule ID	Validation Check	Expected Outcome
V4	discount_percent is between **0 and 100**	No out-of-range values
V5	order_date falls within **Q4 2024 (Oct 1 – Dec 31)**	No dates outside range
V6	payment_method matches one of the **5 defined options**	No undefined methods
V7	If delivery_status = **"Delivered"**, then delivery_date must exist	Cross-field consistency
V8	If return_initiated = **"No"**, then return_reason must be blank	Business logic compliance
V9	warehouse_origin is one of **Mumbai, Delhi, Bangalore**	No other warehouses
V10	refund_status = **"Not Applicable"** when return_initiated = **"No"**	Logical consistency

Cross-Field Consistency Checks:

- For orders with delivery_status = "RTO," verify that return_initiated should logically be "No" (RTO means shipment never reached customer)
- For customer_type = "New," confirm the customer_id doesn't appear in earlier orders (you won't have historical data, but flag if same customer_id appears multiple times with "New" status in this dataset)

Business Logic Validation:

- Calculate average delivery time (delivery_date - order_date). Flag any negative values or values > 30 days
- Verify return window: If return_received - delivery_date > 30 days, flag as exception

Note: We've intentionally included some common data entry errors in this dataset. Your job is to find them and propose how to handle them.

SECTION 3: DATA ANALYSIS TASKS

Complete these in an "Analysis" tab (or multiple tabs if cleaner).

3.1 Descriptive Analysis

Task: Build a Q4 2024 summary dashboard answering:

- Total orders, total revenue (after discounts), total units sold
- Average order value (AOV)
- Overall return rate (returns initiated / delivered orders)
- Breakdown of orders by month

Deliverable: Summary table with clear labels

3.2 Trend Identification

Task: Analyze weekly trends for:

- Order volume
- Return rate
- Average discount percentage applied

Questions to Answer:

- Was there a noticeable spike during Diwali week (Oct 28 - Nov 3)?
- Did return rates lag behind order spikes by 2-3 weeks (as returns take time)?
- Did heavy discounting correlate with higher return rates?

Deliverable: Trend analysis with your interpretation (not just charts)

3.3 Segmentation Analysis

Task: Compare performance across:

Segment A: Customer Type

- New vs. Returning customers: Order volume, AOV, return rate
- Which segment is more profitable after accounting for returns?

Segment B: Order Source

- Website vs. Mobile App vs. Instagram Shop
- Which channel has the best return rate? Highest AOV?

Segment C: Payment Method

- Is COD (Cash on Delivery) associated with higher return/RTO rates compared to prepaid methods?

Segment D: Product Category

- Which category has the highest return rate?
- Is this category also high-volume or is it a smaller, problematic segment?

Deliverable: Comparison tables with your business interpretation

3.4 KPI Calculation

Calculate the following metrics:

KPI	Definition	Your Calculation
Gross Merchandise Value (GMV)	Total order value before discounts	= Σ (order_total)
Net Revenue	GMV minus total discounts	= Σ (order_total) − Σ (order_total × discount_percent / 100)
Return Rate	Proportion of delivered orders with return initiated	= Count(return_initiated = "Yes") / Count(delivery_status = "Delivered")
RTO Rate	Share of orders marked as Return to Origin	= Count(delivery_status = "RTO") / Count(All shipped orders)
Refund Processing Rate	Percentage of refunds successfully processed	= Count(refund_status = "Processed") / Count(return_received not blank)
First-Time Buyer Return Rate	Return rate among new customers	= Returns (customer_type = "New") / Delivered orders (customer_type = "New")
Repeat Buyer Return Rate	Return rate among returning customers	= Returns (customer_type = "Returning") / Delivered orders (customer_type = "Returning")
Fulfillment Split	Distribution of orders by warehouse	= Orders per warehouse_origin / Total orders

3.5 Business Interpretation

Questions to Address:

1. If the overall return rate is 19%, but returning customers have a 10% return rate while new customers have 28%, what does this tell you?
2. If 45% of returns cite "Size Issue," what operational changes might help?
3. If Instagram Shop has the highest return rate, should we shut it down? What else would you want to know before recommending that?

SECTION 4: DATA VISUALIZATION TASK

Create 4-5 visualizations that tell a story. Include these in your PDF submission.

Required Visualizations:

Viz 1: Weekly Order Volume vs. Return Rate (Dual-Axis Line Chart)

- Purpose: Show the relationship between sales spikes and lagged return rates
- Insight to communicate: Whether promotional periods led to quality issues

Viz 2: Return Rate by Product Category (Bar Chart with Benchmark Line)

- Purpose: Quickly identify which categories are problematic
- Include the overall average as a benchmark line

Viz 3: Return Reasons Breakdown (Pie or Donut Chart — only if 5 or fewer categories; otherwise use horizontal bar)

- Purpose: Prioritize which return reasons to address first
- Insight: What % of returns are within our control (Quality Defect, Wrong Item) vs. customer preference (Changed Mind)?

Viz 4: Customer Type × Order Source Performance Matrix (Heatmap or Grouped Bar)

- Purpose: Identify if certain combinations (e.g., New + Instagram) are high-risk
- Show return rate or another relevant metric in the matrix

Viz 5: COD vs. Prepaid — Return and RTO Comparison (Grouped Bar Chart)

- Purpose: Quantify the financial risk of COD orders
- Insight: Should we reduce COD availability for certain segments?

Visualization Guidelines:

- Every chart needs a clear title that states the insight (not just "Return Rate by Category" but "Tops Have the Highest Return Rate at 24%")
- Label axes properly
- Avoid 3D charts, excessive colors, or decoration
- Focus on clarity and business relevance

SECTION 5: PROBLEM-SOLVING / BUSINESS CASE

Scenario:
You've presented your initial findings to the Operations Head. She's now asking for your recommendation on a specific problem:

"Our return rate has jumped from 12% to 19%, and Finance estimates this costs us ₹18-22 lakhs per month in reverse logistics, restocking, and refunds. The CEO wants to get this below 15% within 2 quarters without significantly impacting sales. What do you recommend?"

Your Task:

Write a 1-page recommendation (in your PDF) addressing:

Part A: Root Cause Hypothesis
Based on your analysis, what are the 2-3 most likely drivers of the increased return rate? Support with data.

Part B: Recommended Actions
Propose 3 specific, actionable interventions. For each:

- What you would change
- Which segment/category it targets
- Expected impact (use rough estimates based on your data)
- Any risks or trade-offs

Part C: What You'd Want to Test

If you could run one A/B test or get access to one additional data source, what would it be and why?

Part D: Assumptions & Limitations

What assumptions did you make? What couldn't you answer with this data alone?

Open-Ended Questions (Answer in 2-3 sentences each):

1. A colleague suggests we should just disable returns for all orders under ₹999 to reduce costs. What's your reaction to this idea?
2. The marketing team wants to run another 40% off sale next month. Based on your analysis, what conditions or guardrails would you recommend?
3. If we offered free alterations for size-related returns instead of full refunds, how would you estimate whether this is cost-effective?

FINAL SUBMISSION CHECKLIST

Before submitting, confirm:

- Excel/Sheets file has clearly labeled tabs (Cleaning Log, Validation, Analysis, etc.)
- PDF is 2-3 pages maximum
- All visualizations have clear titles and labels
- Recommendations are specific and tied to your analysis
- You've noted any assumptions or limitations
- You've tracked your total working time

Good luck! We're genuinely excited to see how you approach this problem. There's no single "right" answer we're looking for clear thinking, attention to detail, and someone who can translate data into decisions.